(Please write your Enrollment Number) Enrollment No. 194

End-Term Examination
(CBCS)(SUBJECTIVE TYPE)(Offline)
Programme: <B. Tech.(CSE-AI/ECE-AI), Semester: <6th>
(May, 2026)

Subject Code: BAI 308	Subject: Cloud Computing
Time :3 Hours	Maximum Marks: 60

Note: Q1 is compulsory. Attempt one question each from the Units I, II, III & IV.

Q1		(2.5*8=20)
a.	Discuss the benefits and challenges of cloud computing in detail.	2.5
b.	Analyze the impact of cloud computing on modern IT infrastructure.	2.5
c.	Identify the cloud service model (SaaS, PaaS, IaaS) in each of the following cases and justify your answer briefly: • Gmail • Google App Engine • Amazon EC2	2.5
d.	Distinguish between front-end and back-end components of cloud architecture with suitable examples.	2.5
e.	Explain Software Defined Networking (SDN).	2.5
f.	Distinguish between full virtualization and paravirtualization based on working mechanism.	2.5
g.	Arrange the following in correct order of execution in container-based deployment and explain briefly: Application → Container → Host OS → Hardware	2.5
h.	A system ensures immutability and transparency of transactions without a central authority. Identify the technology used and state one key feature.	2.5
	UNIT I	
Q2	Explain different cloud deployment models in detail. Compare these cloud models based on Cost, Security, Scalability and Control.	(10)
Q3	Explain the trends in computing that led to the emergence of cloud computing. A company is planning to migrate its IT infrastructure to the cloud. Discuss which deployment model would be most suitable under the following conditions: • High security requirements • Cost constraints • Need for scalability	(10)
	UNIT - II	
Q4	Explain the three cloud service models—SaaS, PaaS, and IaaS—in detail. Compare them with respect to functionality, control, and application scenarios. An organization needs complete control over computing resources, including virtual machines, storage, and operating systems. It also requires flexibility to install and manage its own software environment. Identify the most suitable cloud service model and explain its features and advantages.	(10)

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Q5	Discuss the role of major cloud service providers such as Amazon EC2, Google App Engine, and Microsoft Azure. Explain the services offered by each. Explain the challenges and risks associated with cloud computing adoption.	(10)
	UNIT III	
Q6	Client of a data center wants to run multiple operating systems on a single physical server to improve hardware utilization and reduce costs. Identify the concept being used and explain how it works along with its advantages.	(10)
Q7	Explain Network Function Virtualization (NFV). How does it differ from traditional networking approaches?	(10)
	UNIT IV	
Q8	Explain containerization and Docker in detail. Discuss how they improve application deployment and scalability.	(10)
Q9	Discuss emerging trends in cloud computing. Explain how technologies like blockchain are transforming cloud environments. Discuss its architecture, working mechanism, and advantages.	(10)

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